

Waves and Thermodynamics

Question 1 (Conceptual): Two sound waves of the same frequency propagate through air at different temperatures, 0 °C and 30 °C.

Which wave travels faster? Explain using the relationship between temperature and speed of sound in air.

Question 2 (Calculation): A sound pulse travels through helium gas at 20 °C. The speed of sound in helium at this temperature is approximately 1007 m/s. If the pulse travels a distance of 2.5 km:

- (a) Write down the equation relating speed, distance, and time.
- (b) Calculate the travel time of the pulse.
- (c) If the temperature increases to 100 °C and the speed of sound in helium increases by 10%, what is the new travel time over the same distance?